

Kronos Project — Failed-Strategies Postmortem

Every approach we tried, why it died, with the numbers

Updated 2026-06-30 · Testnet / backtest figures throughout — no real capital was risked.

One root cause killed almost everything. At the frequency we traded, the directional signal was always thinner than the cost of trading on it (~15–20 bps round trip). We proved it across a dozen strategies.

Nothing directional survived the multi-regime / placebo / breadth bar. The one genuinely real edge — the **variance risk premium** — exists, but the simple tradeable version loses to crypto's fat tails, and the version that works needs daily delta-hedging we chose not to run. The discipline of refusing to trust the flattering number is what this project actually produced — and it saved real money at every step.

Master scorecard

Strategy	What it was	Best honest result	Verdict
Kronos (core)	Kronos-small ML price predictor	Take-all 47–49% win, Sharpe –0.04 to –0.09	DEAD — never cracked 50%
Kelly / meta-labeler	Logistic P(win) filter over journal features	v1 33.3% vs 61.9% baseline; retrain ~45% vs 51%	DEAD — anti-predictive; never left shadow
Entry-rule search	Mined 2–3 condition combos from harvested trades	4 'greens', OOS 0.07–0.27, never replicated	DEAD — data-mining noise
Consensus TA board	7-indicator trend net-bias	48% win at 1:1 over 5 months	DEAD — lost money with the signal
RSI-2 reversion	Connors fade-the-extreme	No breadth+persistence survivor net of fees	DEAD — didn't clear cost wall
OFI (order-flow)	Cont-style queue-change signal	Net-negative as taker; maker adversely selected	DEAD — edge < spread
Lead-lag	Alt under/over-shoot vs BTC-implied move	Raw lead ≈ 0 (~15s HFT); catch-up failed OOS	DEAD — too fast / no edge
Funding carry	Delta-neutral long spot / short perp	Headline ~19%/yr; our net thin, tail-risky	SET ASIDE — compressing
Pairs / stat-arb	Cointegration mean-reversion	Beat passive on paper; our OOS weak	SET ASIDE — marginal after costs
Vol risk premium (VRP)	Sell overpriced implied vol on Deribit	Premium REAL (proxy ~1.8, implied>realized 74%) but defined-risk condor LOSES (Sharpe –0.04 to –0.8 — fat-tail breaches)	REAL edge, not cleanly harvestable by retail
Crypto-Analyst (as alpha)	Calibrated regime/ensemble to gate or trade	Gates failed placebo (p=0.18/0.36); swing failed breadth (2/7 coins)	DEAD as alpha — its own model card agrees
Intraday-TSM	Morning→afternoon session momentum	+16% 1yr → –17% / Sharpe 0.22 / –52% DD 5yr; selective worse OOS	NOT CLOSED (live only) — backtests failed; live forward test
Stablecoin-flow	On-chain stablecoin supply timing	Sharpe ~1.3 1yr → –8.4% / Sharpe –0.17 5yr vs buy-hold +40%	DEAD — failed 5yr; regime luck

The one genuinely real edge — and why we still couldn't harvest it

The **variance risk premium (VRP)** is the only thing this project couldn't disprove. On BTC, implied vol (Deribit DVOL) sits systematically **above** realized — implied beat realized **74% of days, by ~9.7 vol-points**. The proxy backtest (a daily-marked variance swap) showed **Sharpe ~1.8**, positive in both halves, surviving cost stress and beating ~96% of placebo shuffles. Unlike every directional signal it didn't reshuffle — because it isn't a prediction. You're paid to bear tail risk, like an insurer. That premium is real.

But the tradeable version failed. We modeled the actual trade — a monthly defined-risk iron condor, priced on real DVOL with the real capped payoff. Across strikes and tenors it came out **net-negative (Sharpe -0.04 to -0.8)**, an 82–91% win rate masking a few -100% capped tails that wiped out the small wins. The tell: even priced at *realized* vol (no premium), the structure loses ~-5%/cycle — because **crypto's endpoints are fatter-tailed than daily vol predicts**, so a static short structure held to expiry gets run over by jumps. The +2.5%/cycle premium couldn't cover the fat-tail drag.

The reconciliation: the ~1.8 proxy is a *continuously delta-hedged* variance swap — it harvests implied-minus-realized variance regardless of the endpoint. The condor is a *static endpoint bet* that the tail kills. The trade that monetizes the premium is a **delta-hedged short straddle** (rehedge daily) — the operationally-intensive, gap-risky institutional version, which we **chose not to run** (a small trader selling vol they aren't confident in is exactly how blow-ups happen). VRP lands in its own category: a real edge a small retail trader can't cleanly harvest.

The Simons lens — why fast retail prediction is structurally impossible

Medallion's famous '66%' is its annual *return*, not its win rate (~50.75%). They won on a tiny per-trade edge × ~300,000 trades/day × ~12.5x leverage × near-zero cost. Our cost (15–20 bps) is 10–20x the fast edges we chased — so fast directional scalping is closed to us by arithmetic. The only edges that can work are slow, fatter-per-trade, structural. VRP was the one that fit — and even it doesn't harvest cleanly in retail, defined-risk form.

The honest meta-lesson

The most reliable tell across everything: the **in-sample → out-of-sample collapse**, and the **proxy → tradeable collapse**. Rule search 86%→53% win. Intraday +16% cherry year → -17% over five. Stablecoin Sharpe 1.3 → -0.17. Analyst gate 'OOS 1.0' → placebo p=0.18. VRP proxy 1.8 → tradeable negative. Every time we got excited it was a small sample, an in-sample fit, or a frictionless proxy; every time we demanded honest OOS / placebo / breadth / real-friction evidence, the edge evaporated. **Refusing to trust the flattering number is the most valuable thing this project produced** — it kept real money safe through every one of these.

Internal sources: Kronos_Search_Runs_Tracker.md · CAPTAINS_LOG.md · Creative_Edges_Deep_Dive.md · vol_premium_backtest.py, vrp_structure_backtest.py · analyst_swing_backtest.py, analyst_gate_verify.py · intraday_tsm_equity.py, stable_flow_strategy.py (methods + verdicts).